MONASH University

FIT3181/5215 Deep Learning

ViT and Model Fine-tuning with Prompts, LoRA, and Adapters

Teaching team

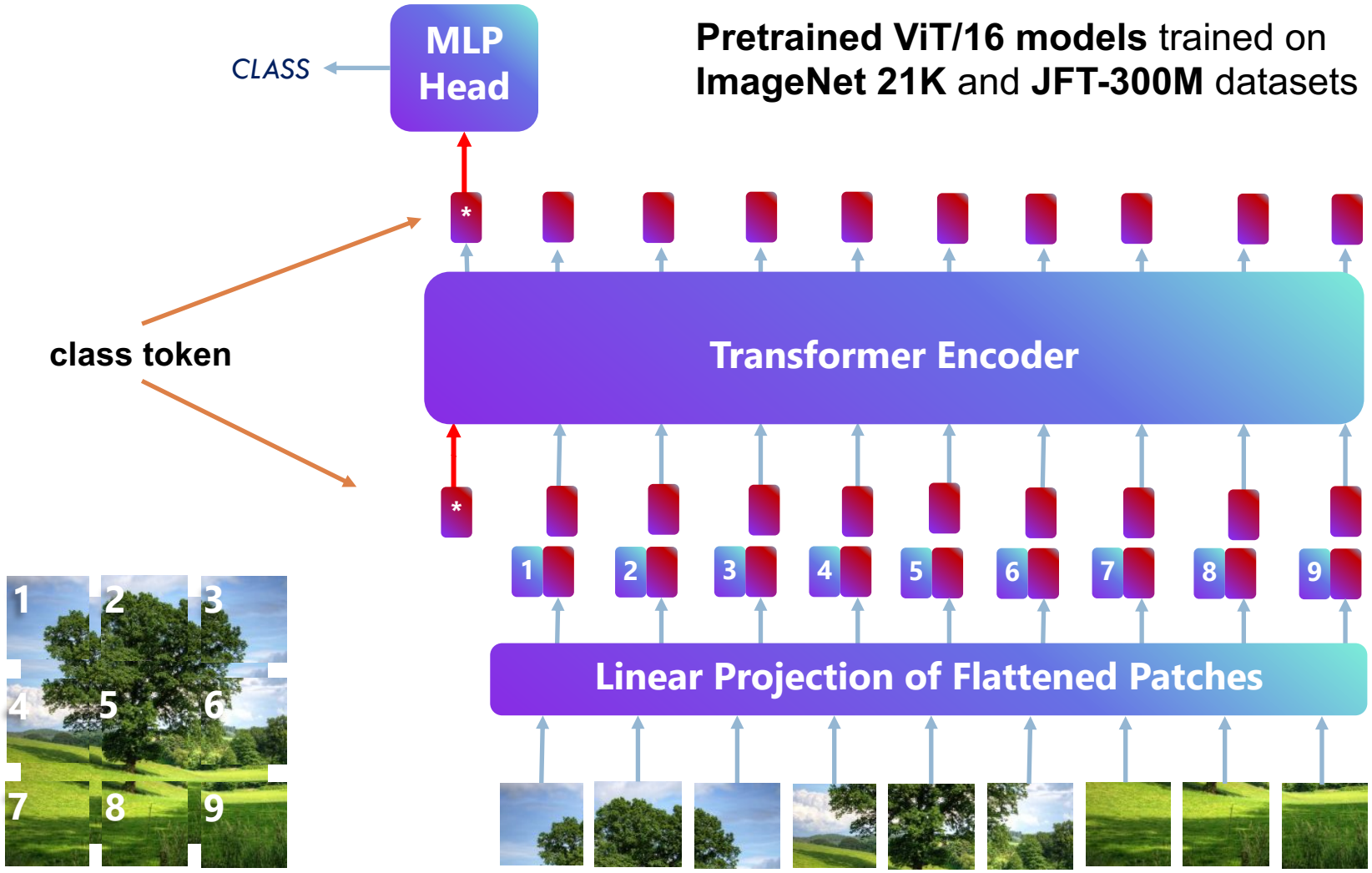
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GROUP OF EIGHT AUSTRALIA

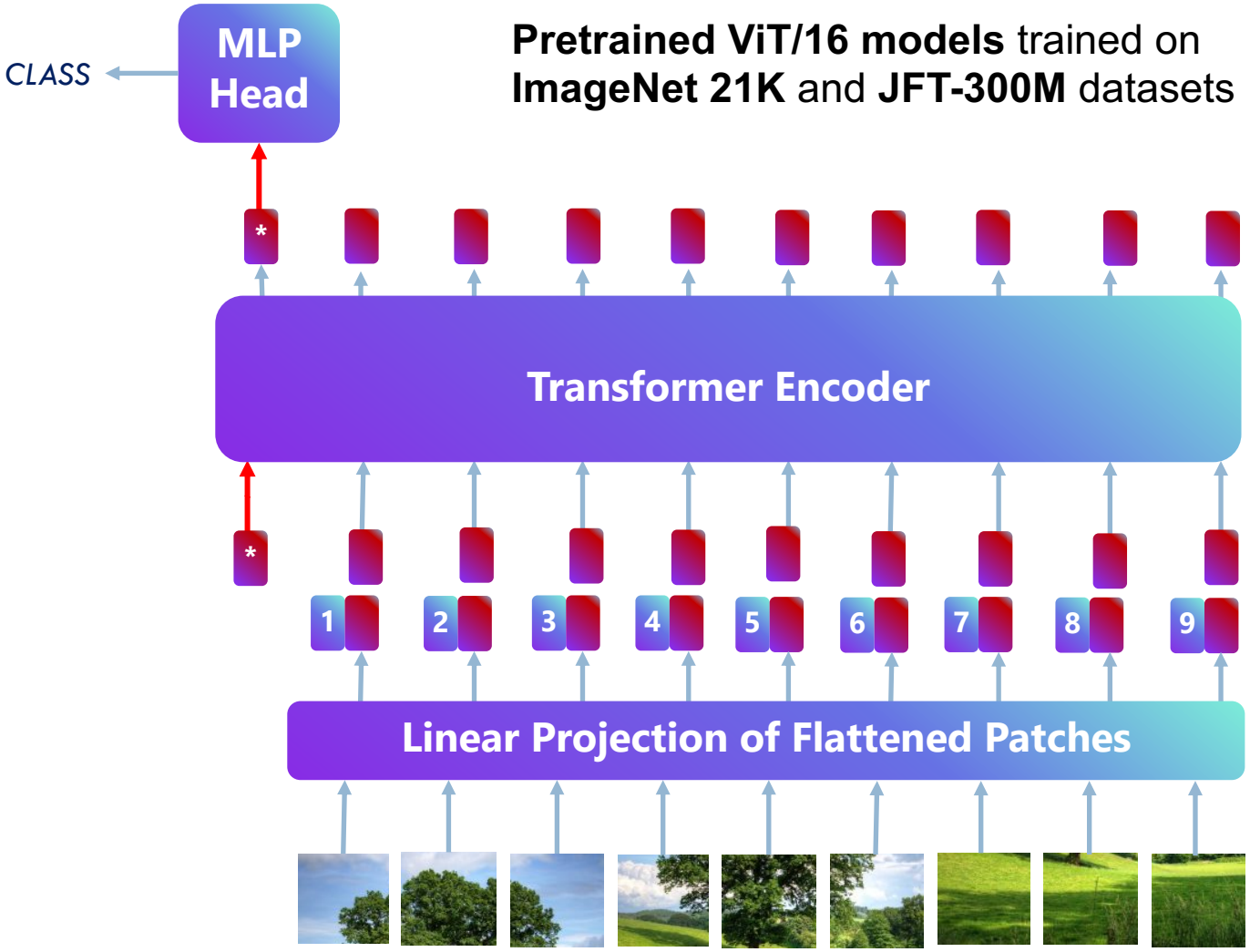
Outline

- Tutorial 10: Model Fine-tuning for ViT

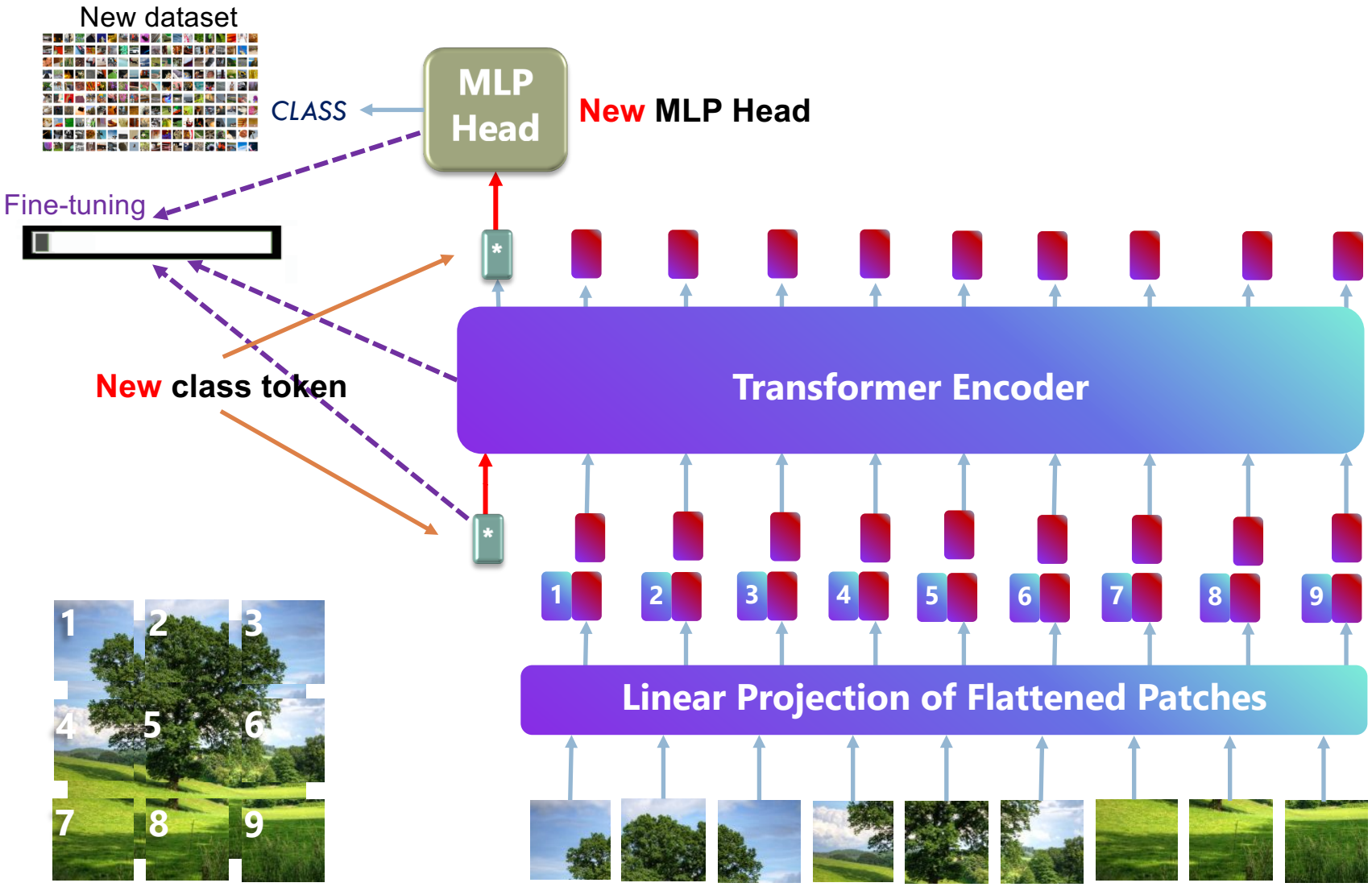
ViT: Model Fine-Tuning



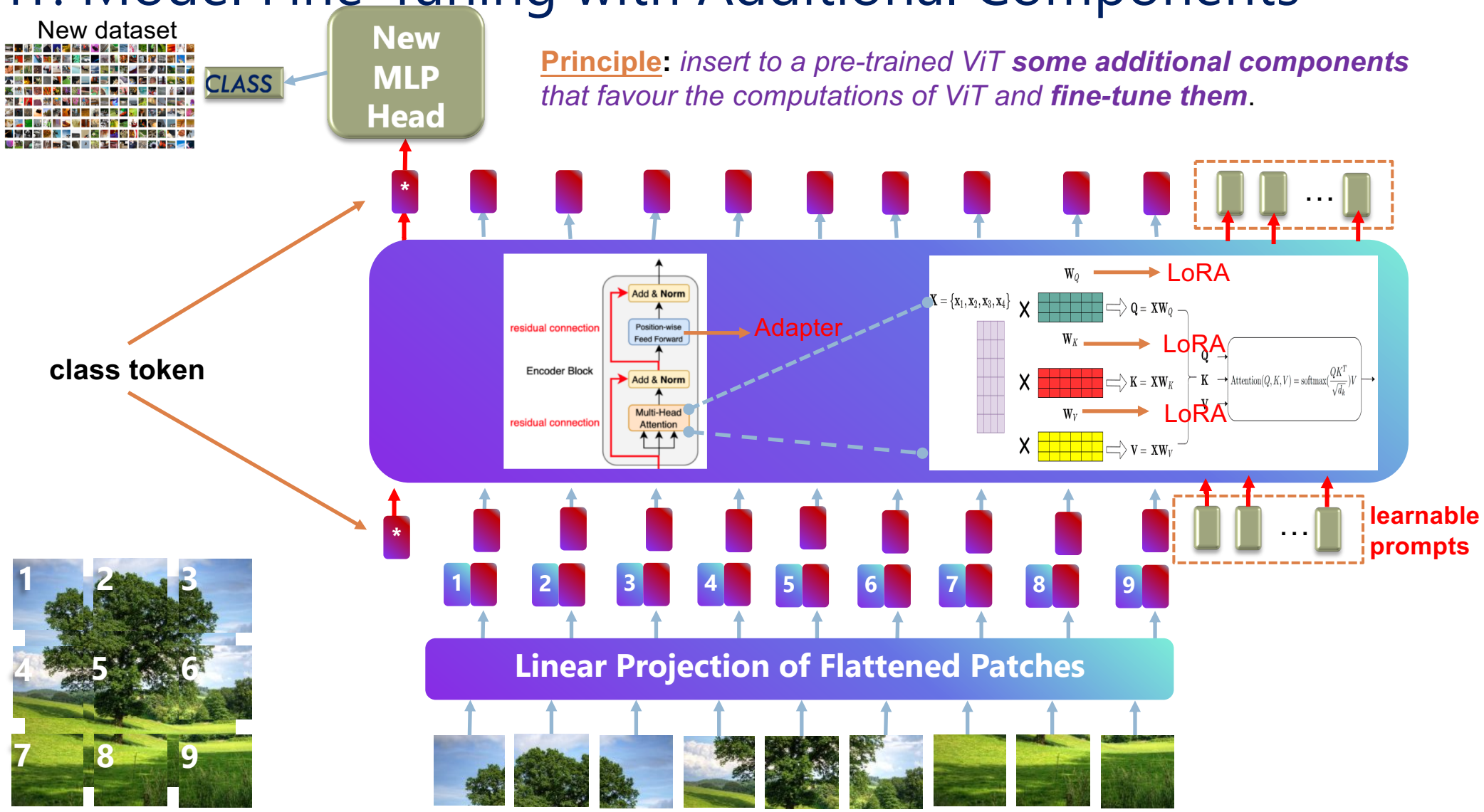
ViT: Model Fine-Tuning



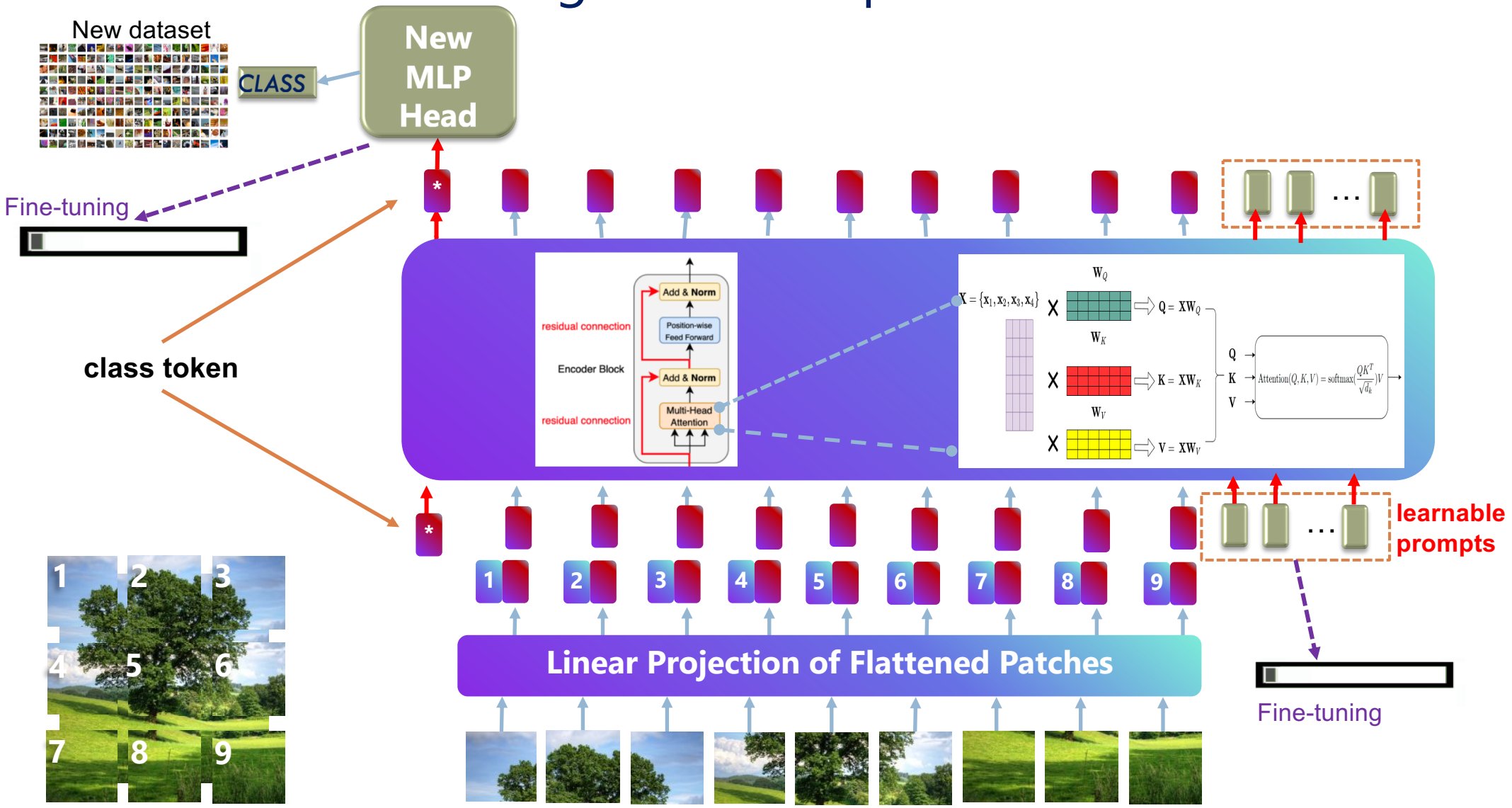
ViT: Model Fine-Tuning



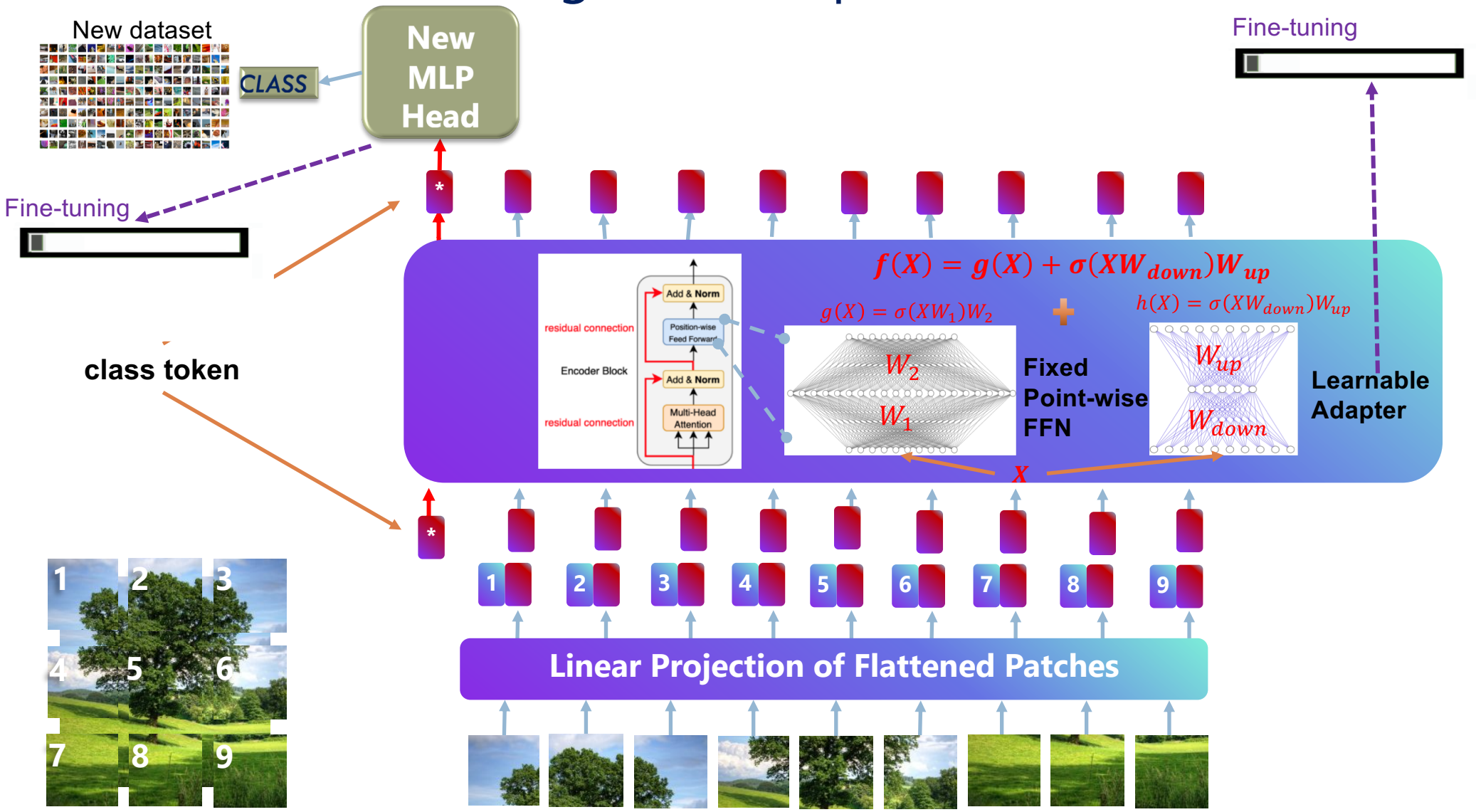
ViT: Model Fine-Tuning with Additional Components



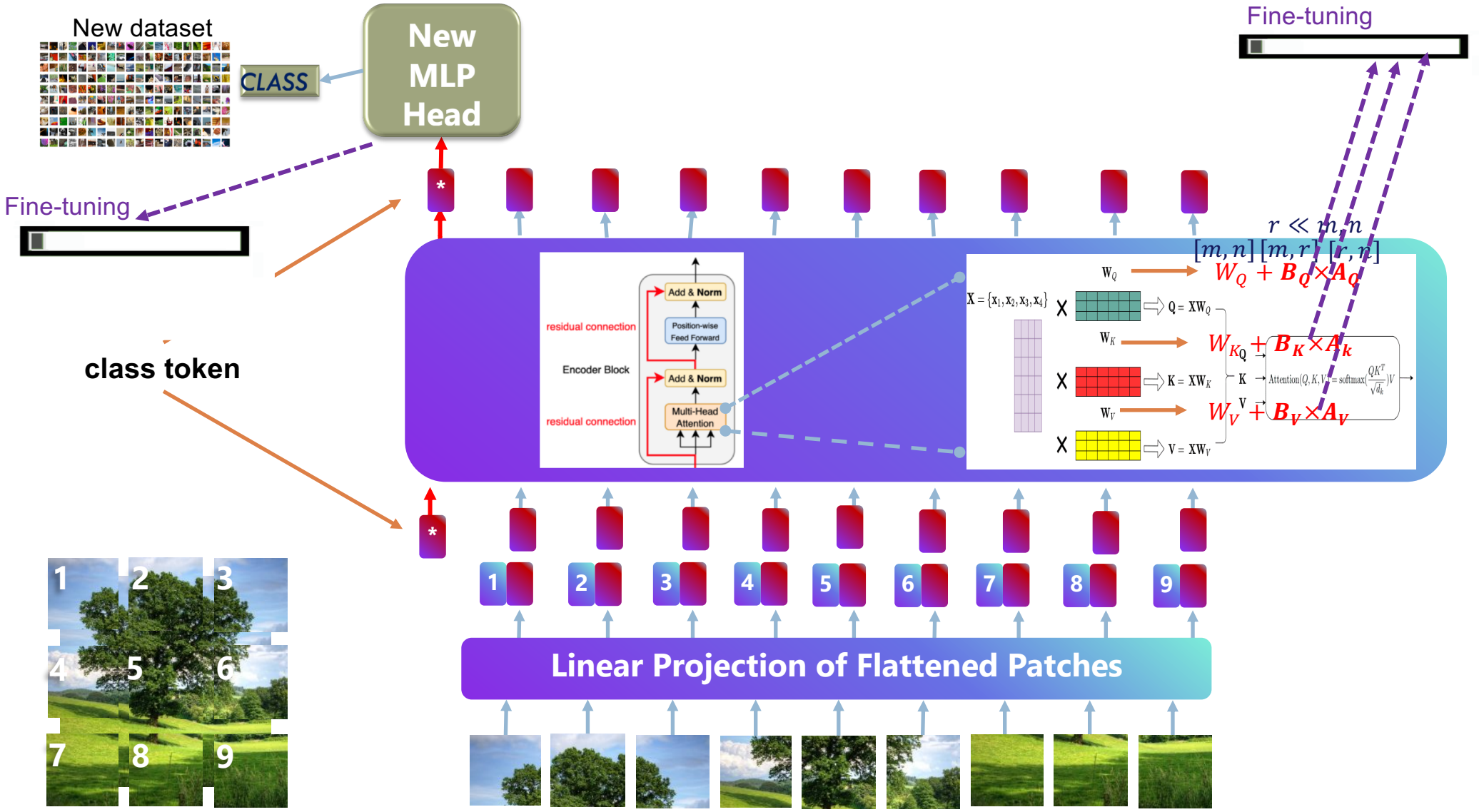
ViT: Model Fine-Tuning with Prompts



ViT: Model Fine-Tuning with Adapters

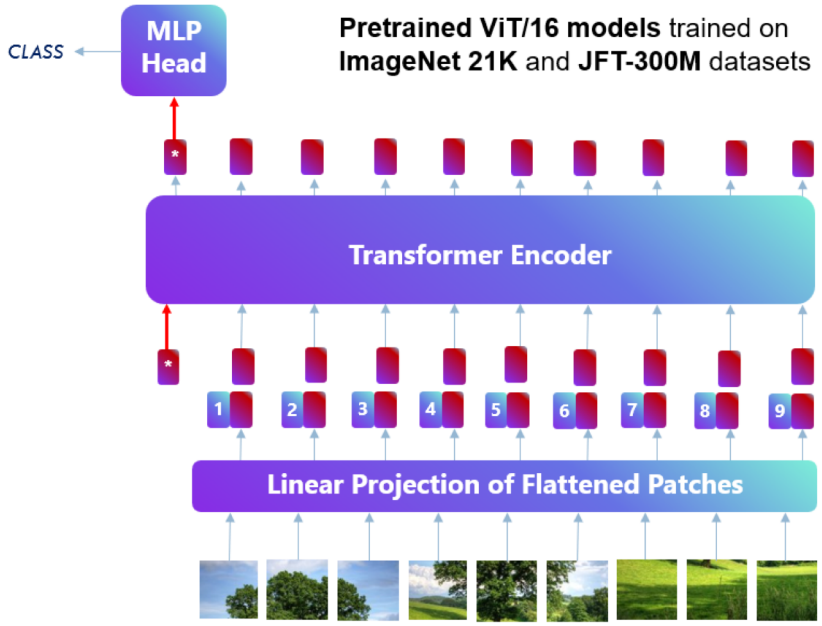


ViT: Model Fine-Tuning with Additional LoRA

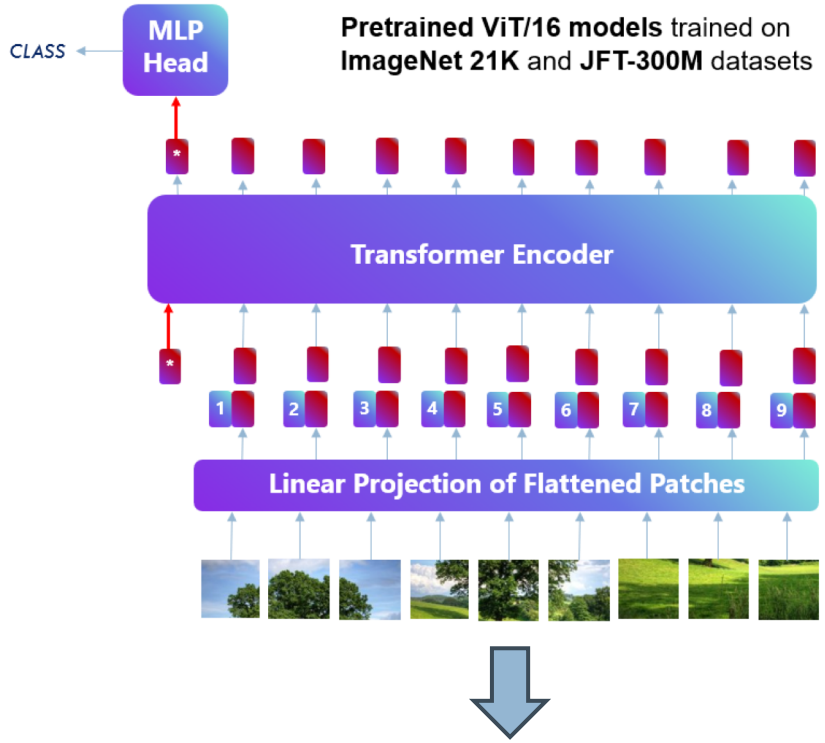


The skeleton of the code

```
# Load imagenet21k pre-train weights for ViT-B_16
!wget https://storage.googleapis.com/vit_models/imagenet21k/ViT-B_16.npz
```



Declare the **architecture of ViT-B_16** and **load the weights and parameters** from the pre-trained model → **our ViT-B_16 in memory**



Base on our ViT-B_16 in memory to do model fine-tuning or model fine-tuning with **prompts** or **adapters**

Thanks for your attention!